Module 2: Fluid Kinematics

1. Introduction to Kinematics

2. Methods of Describing Fluid Motion
   a) Lagrangian Method
   b) Eulerian Method

3. Flow Patterns
   a) Stream Line
   b) Path Line
   c) Streak Line
   d) Streak Tube

Fluid Kinematics

Definition:
It is a branch of science that deals with the motion of fluid without considering forces causing the motion.

The study of **Kinematics** is often referred to as the geometry of motion.

It is generally a continuous function in space time.

Motion Characteristics:
- Velocity
- Acceleration
- Pressure
- Density

Methods of Describing Fluid Motion

Fluid motion is described by two methods.

Methods:

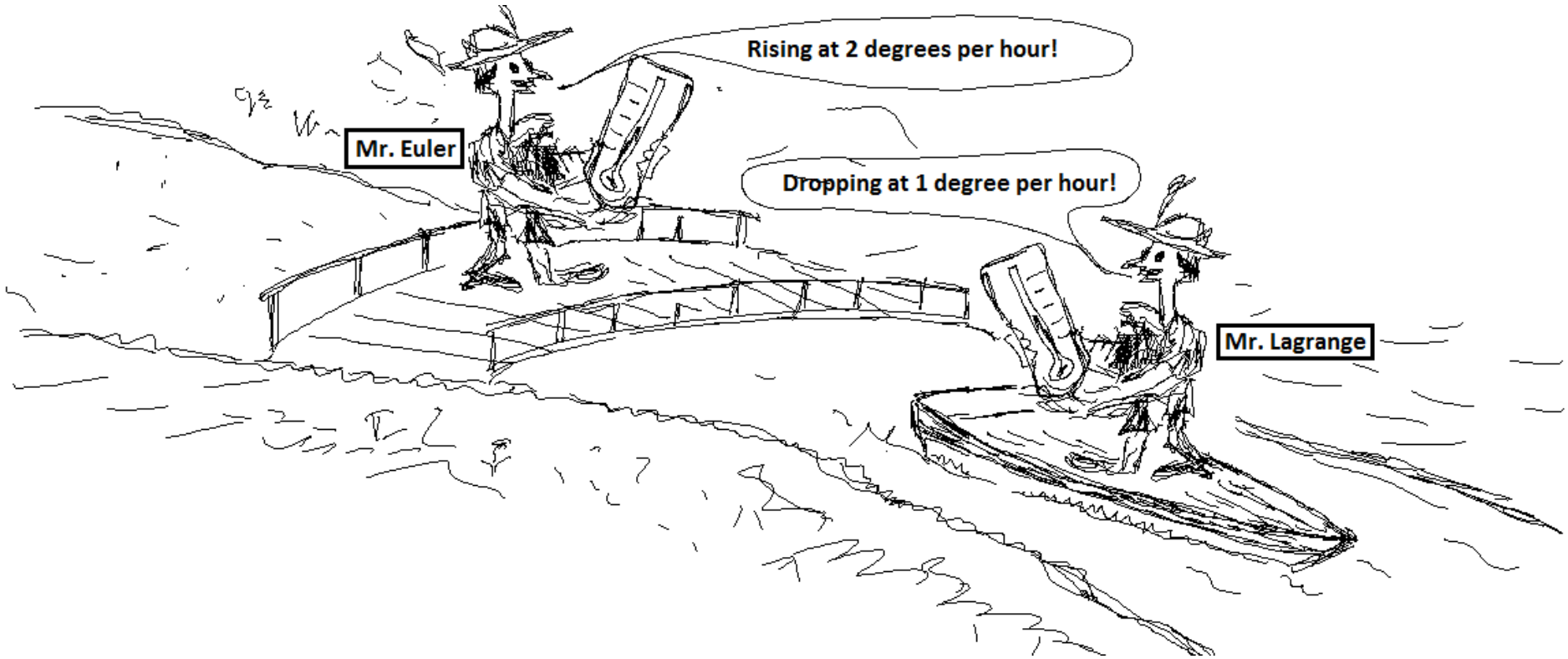
- **Lagrangian Method** – Describes a defined mass (position, velocity, acceleration, pressure, temperature etc.,) as functions of time.

  Example:
  Track the location of a migrating bird

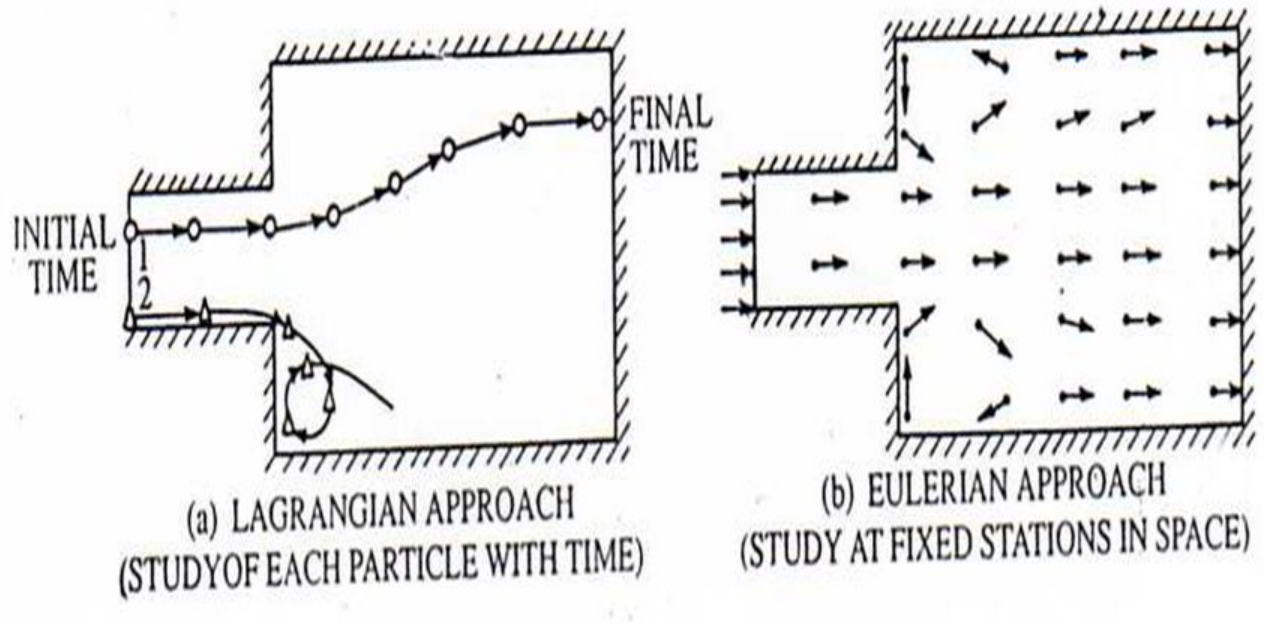
- **Eulerian Method** - Describes a flow field (velocity, acceleration, pressure, temperature etc.,) as functions of position and time.

  Example:
  Count the birds passing through a particular location




Rising at 2 degrees per hour!
Mr. Euler
Dropping at 1 degree per hour!
Mr. Lagrange

- In the **Eulerian approach**, the fluid motion at all points in the flow field is determined by applying the laws of mechanics at all fixed stations.
- This is considerably easier than the **Lagrangian approach** and is followed in the study of Fluid Mechanics.



Difference Between Lagrangian and Eulerian Description

- Imagine a person standing beside a river measuring its properties. In the Lagrangian approach, he throws in a probe that moves downstream with the water. In the Eulerian approach, he anchors the probe at a fixed location in the water.
- Experimental measurements are more suitable to the Eulerian Description.
- Equations of motion of fluid flow in Lagrangian description are well defined (Newtons' second law), but needs to be carefully derived for the Eulerian description

Lagrangian method

- In this method, a single fluid particle is chosen and followed during its motion.
- Its velocity, acceleration, density etc. is described with respect to its location in space and time from a fixed position at the start of the motion.
- The position of the fluid particle (x,y,z) at any time t with respect to its position (a,b,c) at time t=0 is given as,

$x=f1(a,b,c,t)$; $y=f2(a,b,c,t)$; $z=f3(a,b,c,t)$

Then, velocity is given by

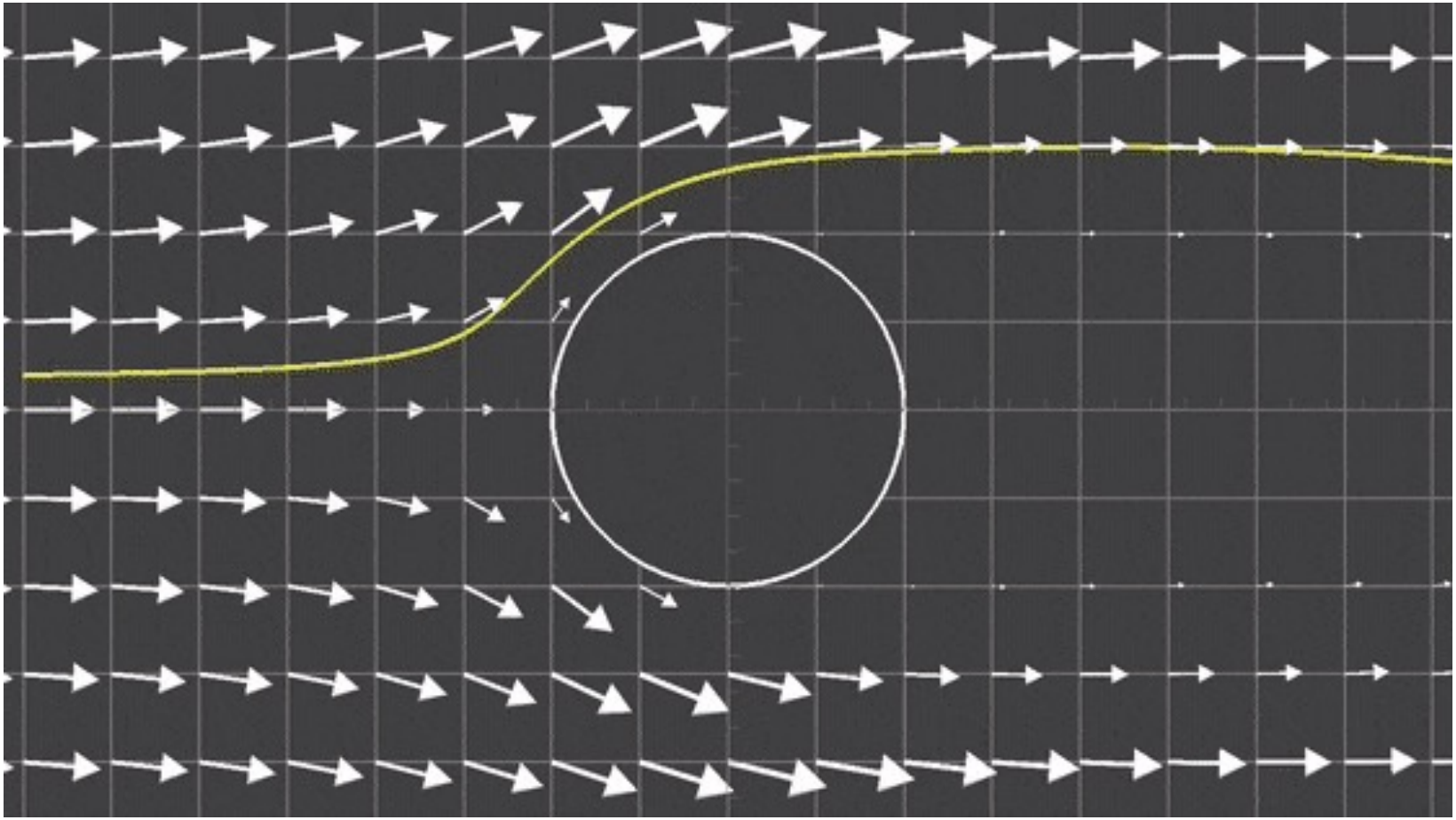
$u=dx/dt$; $v=dy/dt$; $w=dz/dt$

$a_x = du/dt = d^2x/dt^2$; $a_y = d^2y/dt^2$; $a_z = d^2z/dt^2$

Resultant velocity=?

Resultant acceleration=?


Eulerian Frame
Lagrangian Frame



Eulerian method

- In this method, a point or section is chosen in the flow field
- Its velocity, acceleration, density etc. are observed at that point
- It is a commonly used method because of mathematical simplicity
- Let u, v, and w be the components of resultant velocity in x, y and z axis respectively. The velocity components vary along space and time.

$u=f1(x,y,z,t); v=f2(x,y,z,t); w=f3(x,y,z,t)$

Then, resultant velocity is given by

Resultant velocity=?

Flow Visualization

Flow Visualization is the Visual Examination of Flow-Field Features.

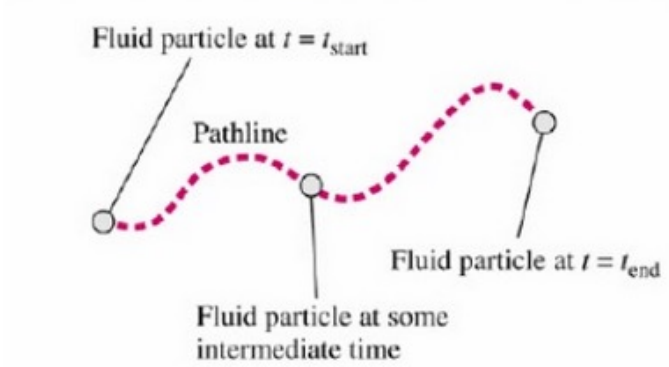
It is important for both physical experiments and numerical solutions.

Description of Flow Patterns:

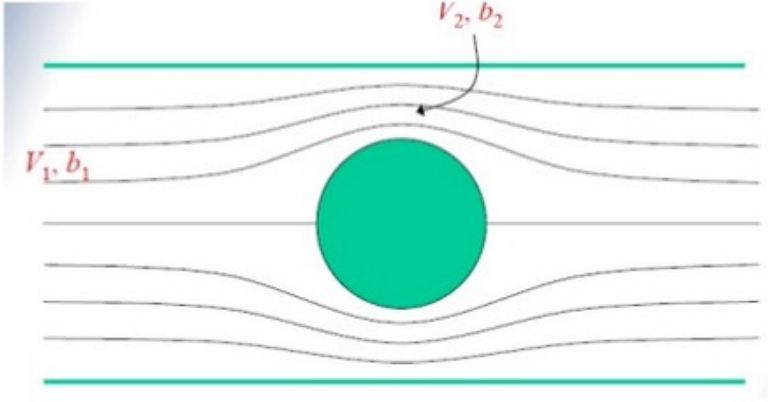
1. Stream Lines
2. Path Lines
3. Streak Lines
4. Stream Tube

Path line and stream line

- **Pathline:** It is trace made by single particle over a period of time.
- **Streamline** show the mean direction of a number of particles at the same instance of time.



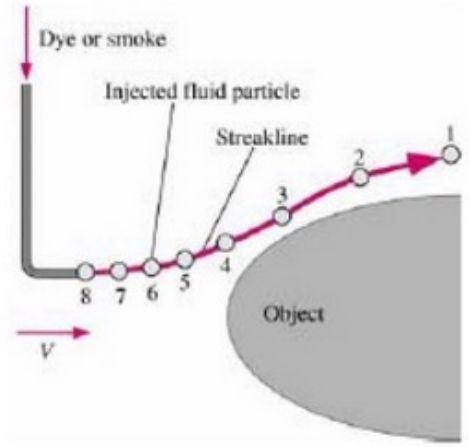
- **Character of Streamline**
- 1. Streamlines can not cross each other. (otherwise, the cross point will have two tangential lines.)
- 2. Streamline can't be a folding line, but a smooth curve.
- 3. Streamline cluster density reflects the magnitude of velocity. (Dense streamlines mean large velocity; while sparse streamlines mean small velocity.)



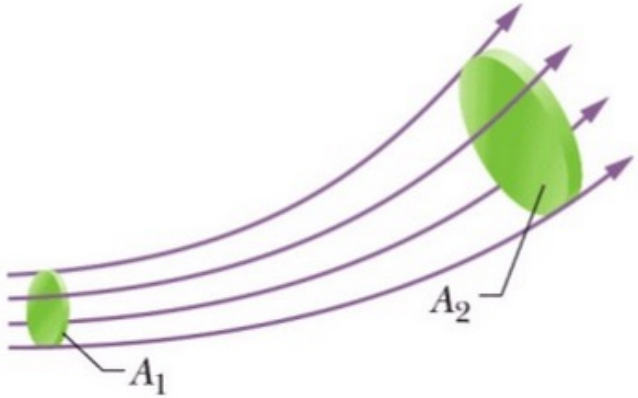
Flow around cylindrical object

Streakline and streamtubes

- A **Streakline** is the locus of fluid particles that have passed sequentially through a prescribed point in the flow.
- It is an instantaneous picture of the position of all particles in flow that have passed through a given point.



- **Streamtube** is an imaginary tube whose boundary consists of streamlines.
- The volume flow rate must be the same for all cross sections of the stream tube.



Types of Fluid Flow

Types of Fluid Flow:

1. Steady and Unsteady Flow
2. Uniform and Non-Uniform Flow
3. Laminar and Turbulent Flow
4. Rotational and Irrotational Flow
5. Compressible and Incompressible Flow
6. Ideal and Real Flow
7. One, Two and Three Dimensional Flow

Steady and Unsteady Flow

Steady flow is that type of flow in which fluid parameters (velocity, pressure, density etc.) at any point in the flow field do not change with time.

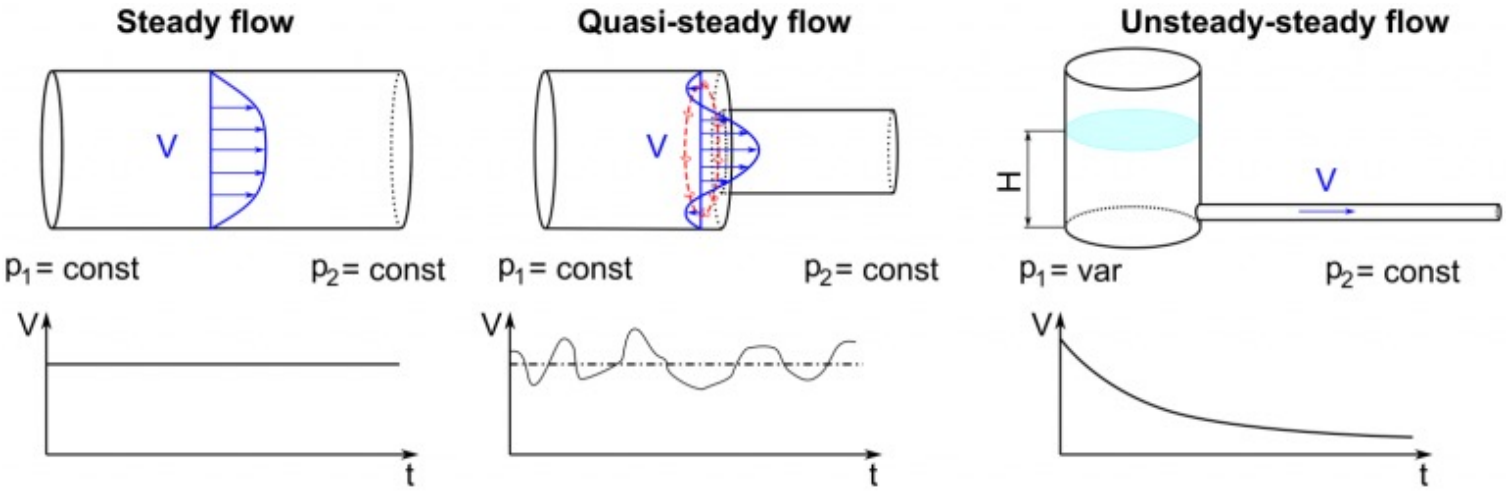
This means that the fluid particles passing through a fixed point have the same flow parameters like velocity, pressure, surface tension etc. The parameters may be different at the different cross-section of the flow passage.

$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} = 0, \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} = 0, \left(\frac{\partial \rho}{\partial t}\right)_{x_0, y_0, z_0} = 0$$

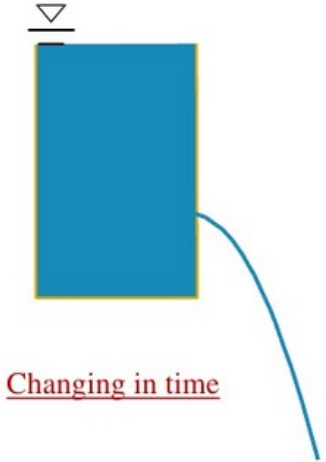
where (x_0, y_0, z_0) is a fixed point in fluid field.

Unsteady flow is that type of flow in which fluid parameters (velocity, pressure, density etc.) at a point changes with time.

$$\left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} \neq 0, \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} \neq 0 \text{ etc.}$$



A flow is said to be **quasi steady** if temporal variations at a spatial location are much smaller (they would be zero if the flow was steady) compared to spatial variations for any quantity.



Uniform and Non-Uniform Flow

Uniform and Non-uniform Flows. Uniform flow is defined as that type of flow in which the velocity at any given time does not change with respect to space (*i.e.*, length of direction of the flow). Mathematically, for uniform flow

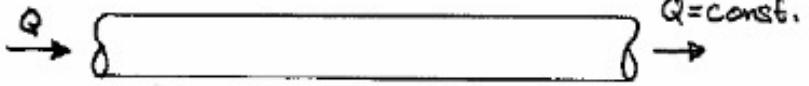
$$\left(\frac{\partial V}{\partial s}\right)_{t = \text{constant}} = 0$$

where ∂V = Change of velocity

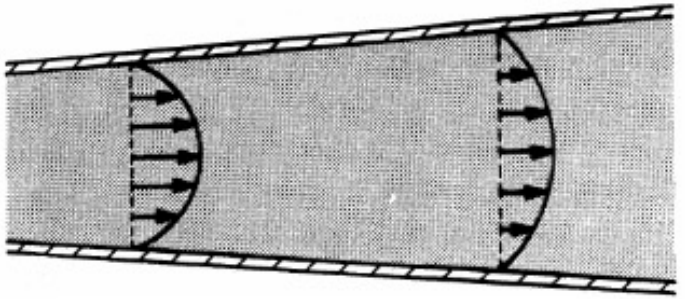
∂s = Length of flow in the direction S.

Non-uniform flow is that type of flow in which the velocity at any given time changes with respect to space. Thus, mathematically, for non-uniform flow

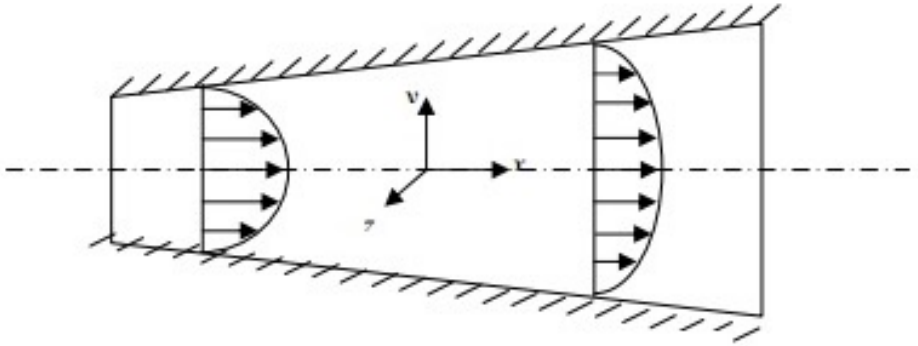
$$\left(\frac{\partial V}{\partial s}\right)_{t = \text{constant}} \neq 0.$$



Steady uniform flow



Steady non-uniform flow



Three dimensional flow

Laminar and Turbulent Flow

Laminar flow is also called streamline or viscous flow. This type of flow occurs in smooth pipes having the low velocity of flow. It also occurs in liquids having high viscosity.

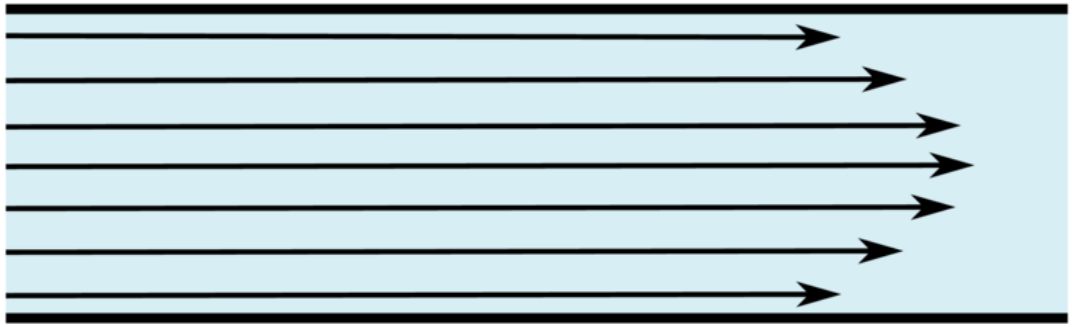
Turbulent flow is defined as that type of flow in which each fluid particle does not have a definite path and the paths of individual particles cross each other.

In other words, it is the flow in which fluid particles move in a zigzag path

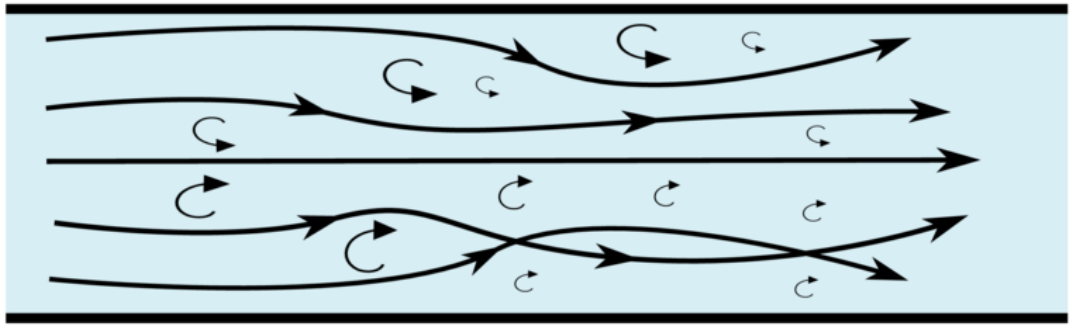
When a fluid is flowing in a pipe, the type of flow is determined by a non-dimensional number, called **Reynold's number**.

For laminar flow, Reynold number $<$ 2000
For turbulent flow, Reynold number $>$ 4000

laminar flow

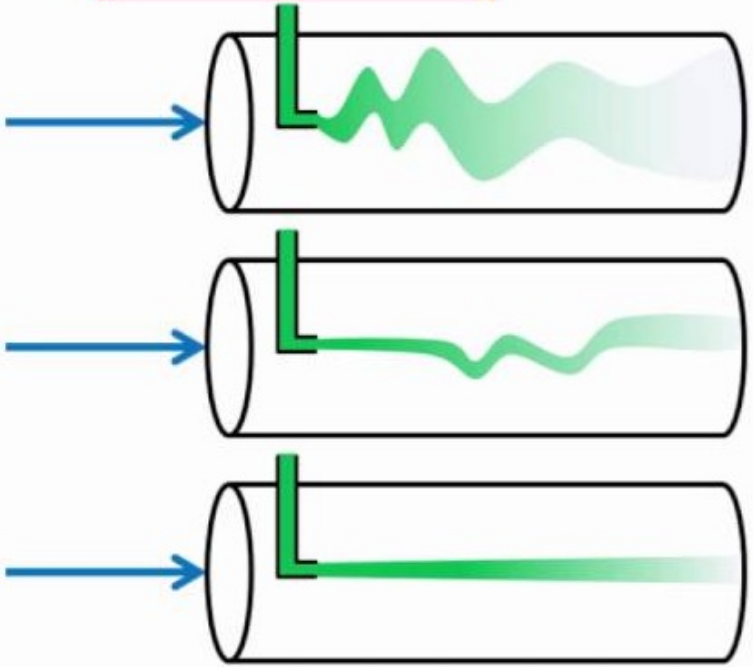


turbulent flow



The Reynolds number correlates well with flow characteristics.

$$\mathrm{Re} = \frac{\rho V_{\mathrm{avg}} D}{\mu}$$



$\mathrm{Re} > 4000$

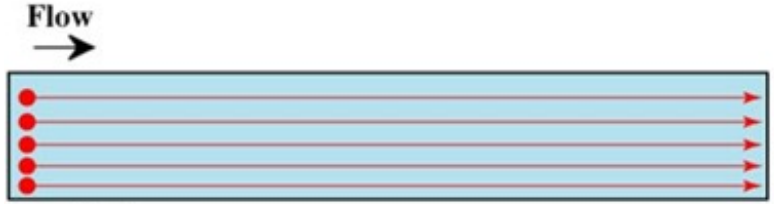
turbulent (unpredictable, rapid mixing)

$2300 < \mathrm{Re} < 4000$

transitional (turbulent outbursts)

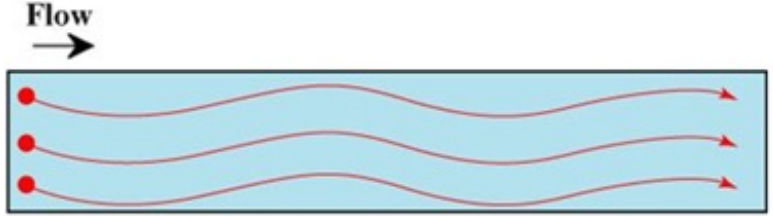
$\mathrm{Re} < 2300$

laminar (predictable, slow mixing)



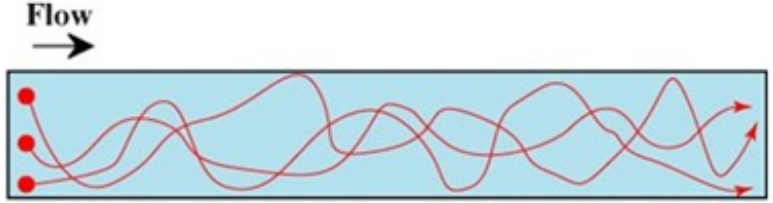
Low discharge

Laminar Flow: every fluid molecule followed a straight path that was parallel to the boundaries of the tube.



Medium discharge

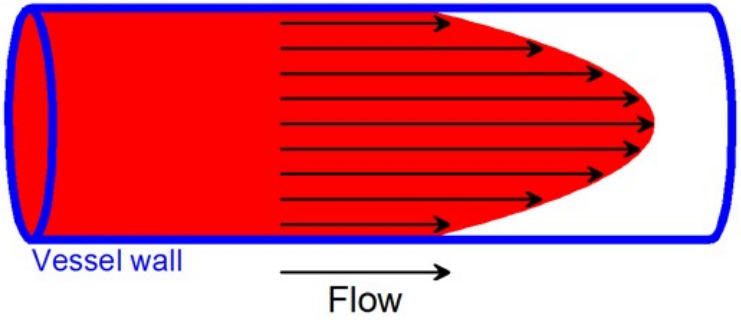
Transitional Flow: every fluid molecule followed wavy but parallel path that was not parallel to the boundaries of the tube.



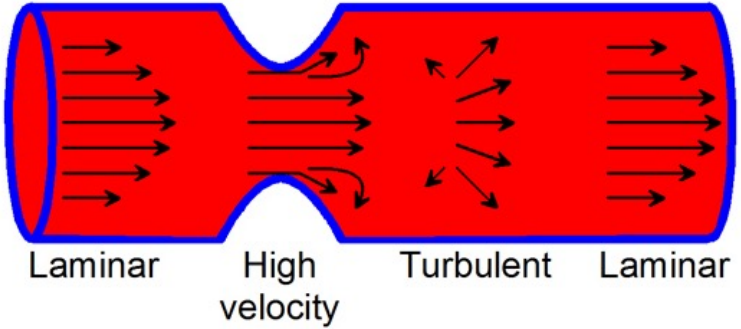
High discharge

Turbulent Flow: every fluid molecule followed very complex path that led to a mixing of the dye.

Laminar blood flow



Turbulent blood flow





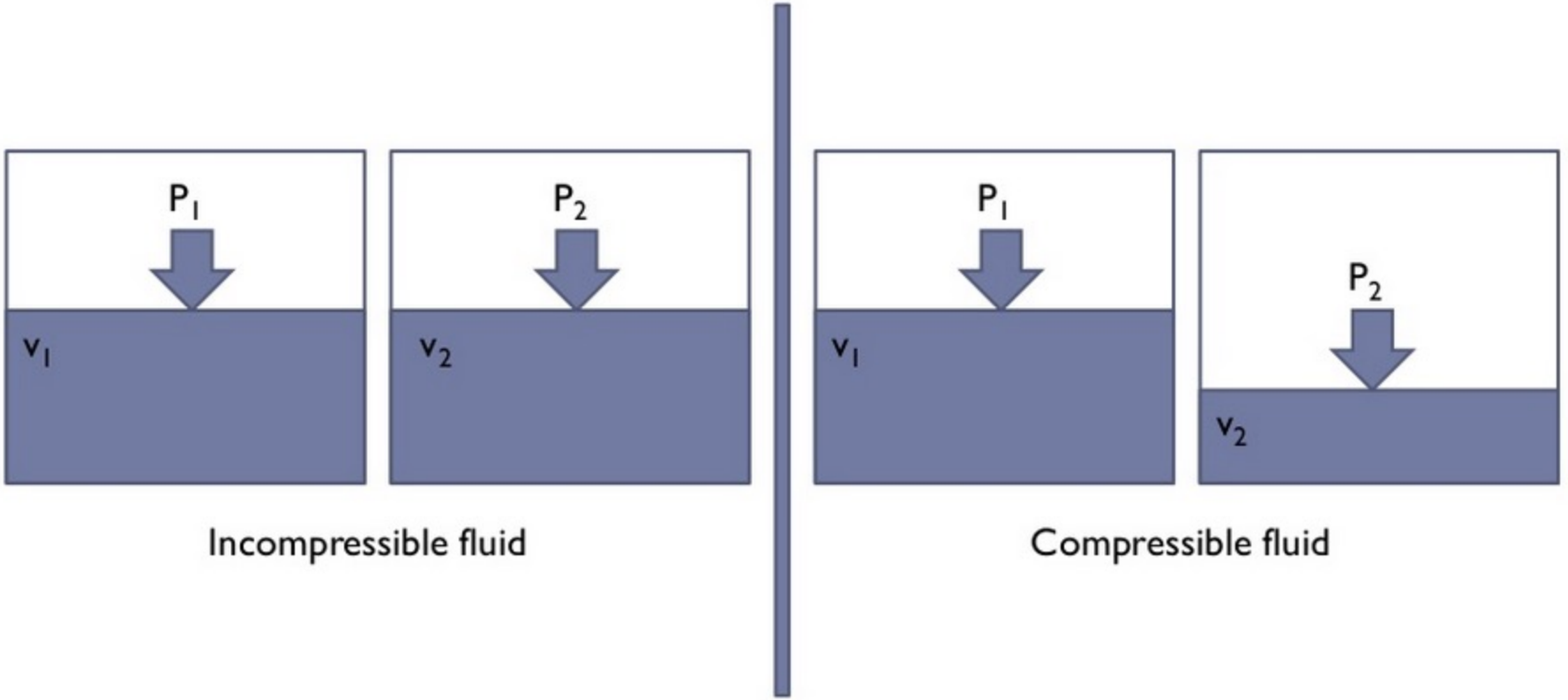
Compressible and Incompressible Flow

The flow in which the density of fluid changes, due to pressure and temperature variations, from point to point during the flow is called **compressible flow**. In other words, it is the flow in which the density of a fluid is not constant during the flow.

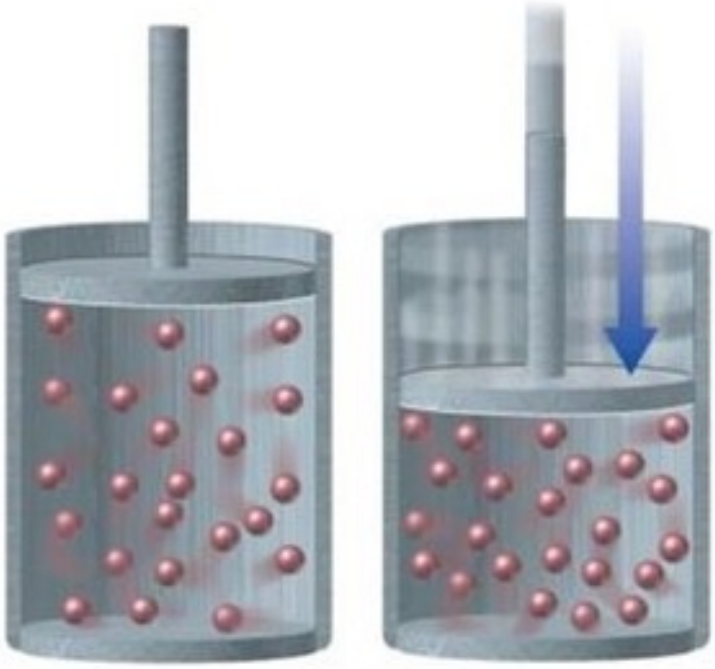
Mathematically, for compressible flow, $\rho \neq$ constant

The flow in which the density of fluid does not change during the flow is called **incompressible flow.** In other words, it is the flow in which the density of a fluid is constant during the flow.
Mathematically, for incompressible flow, $\rho =$ constant

Liquids are generally incompressible which means that pressure and temperature changes have a very little effect on their volume. Gases are compressible fluids.

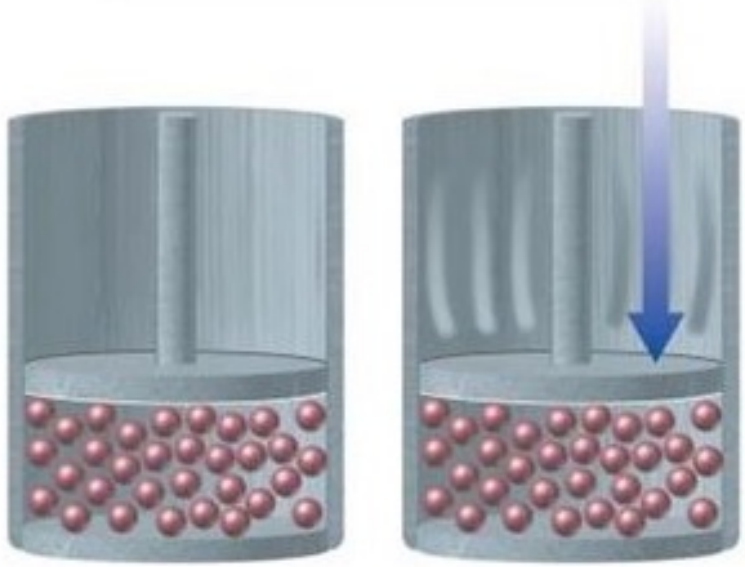

P1
P2
v1
v2
P1
P2
v1
v2
Incompressible fluid
Compressible fluid

Gases are compressible.



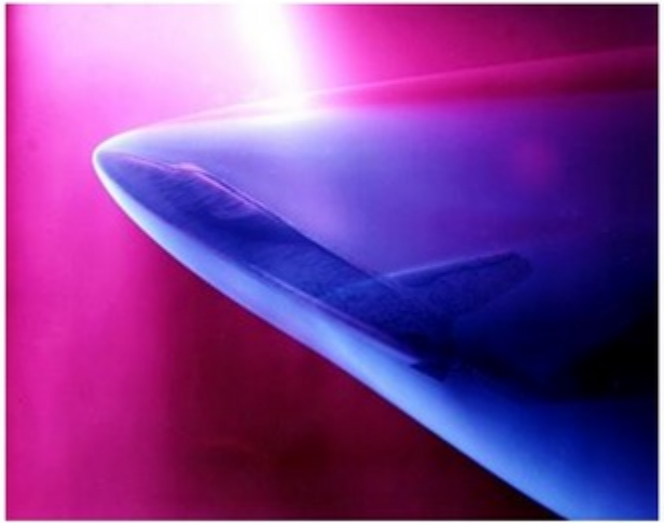
Gas

Liquids are not compressible.



Liquid

Compressible Flow

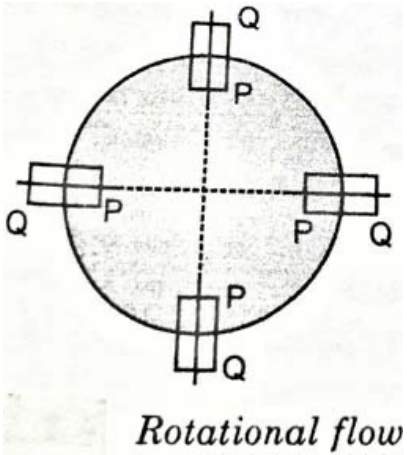




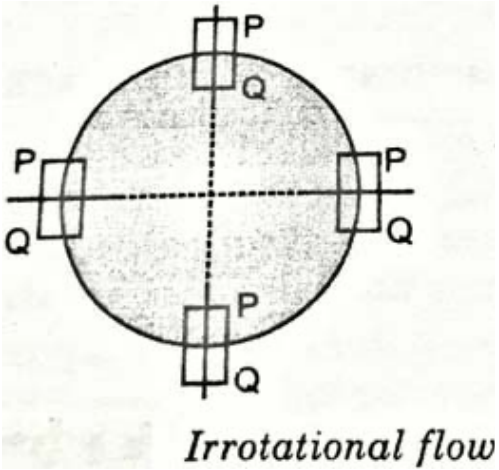
Rotational and Irrotational Flow

Rotational flow is that type of flow in which fluid particles also rotate about their own axes while flowing along a streamline.

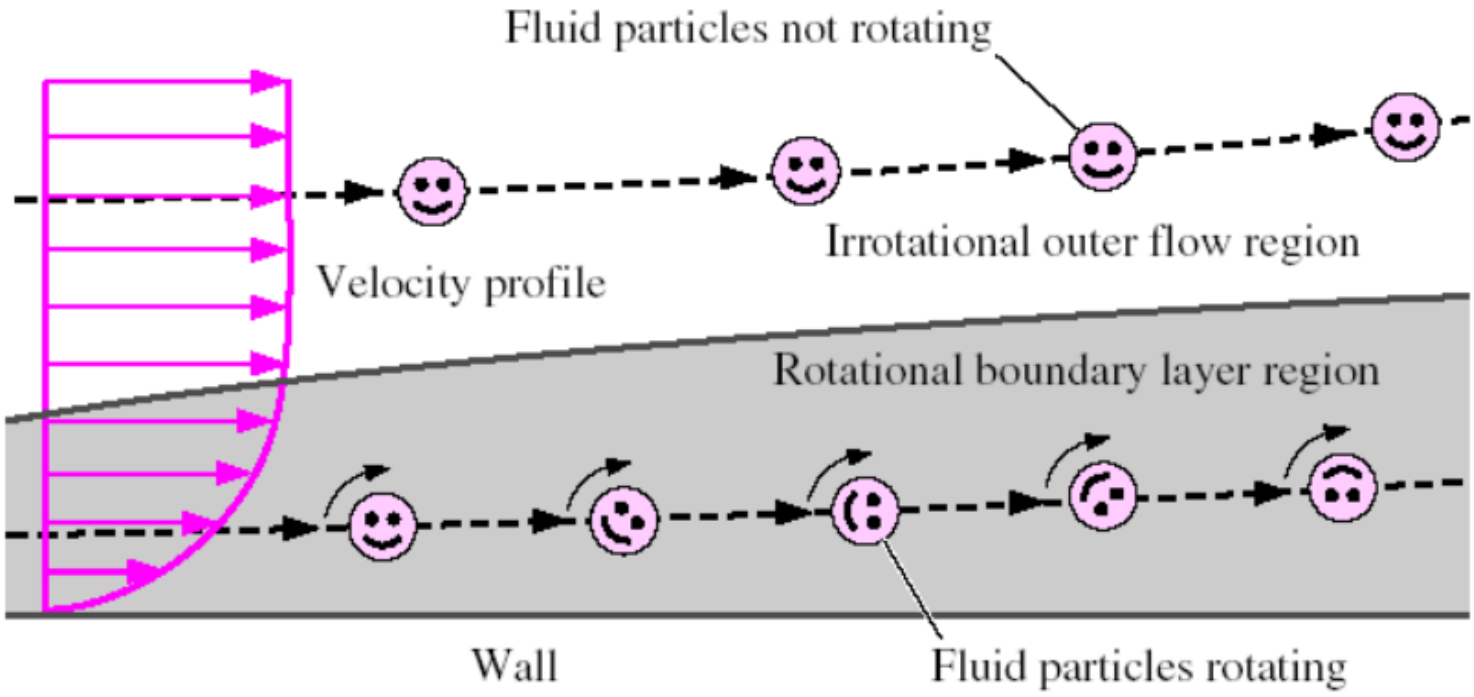
Irrotational flow is that type of flow in which fluid particles do not rotate about their own axes while flowing.

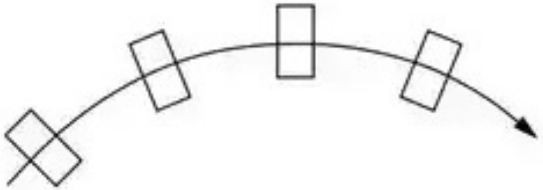


Rotational flow

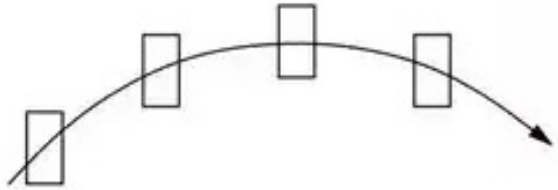


Irrotational flow

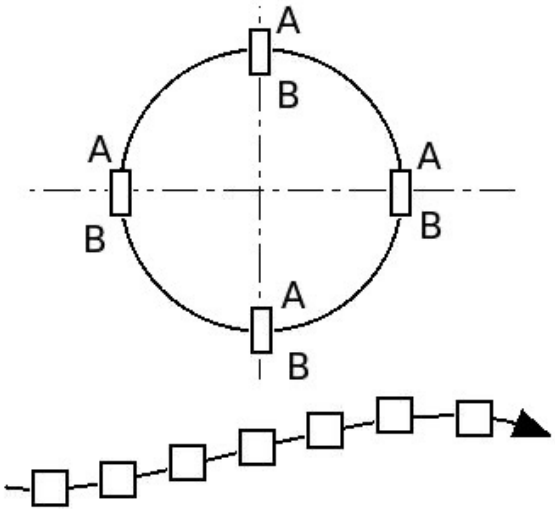

Fluid particles not rotating
Irrotational outer flow region
Velocity profile
Rotational boundary layer region
Wall
Fluid particles rotating



Rotational flows



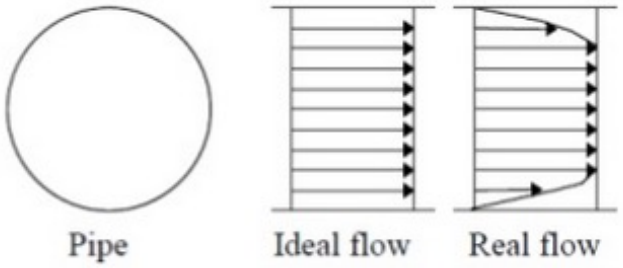
Irrotational flows



Ideal and Real Flow

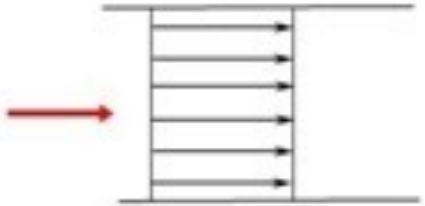
An **ideal flow** is the flow of a non-viscous fluid. In the ideal flow, no shear stress exists between two adjacent layers or between the fluid layer and boundary, only normal stresses can exist in ideal flows.

The flow of **real (viscous) fluids** is called real flow. In real flow, shear stress exists between to adjacent fluid layers. These stresses oppose the sliding of one layer over another.



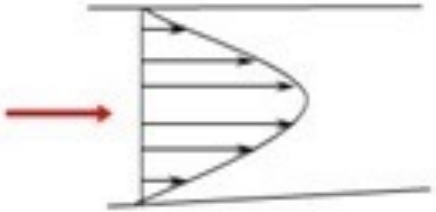
Velocity distribution of pipe flow

Real fluid flows implies friction effects. Ideal fluid flow is hypothetical; it assumes no friction.



Ideal

Friction = 0
Ideal Flow (μ =0)
Energy loss =0



Real

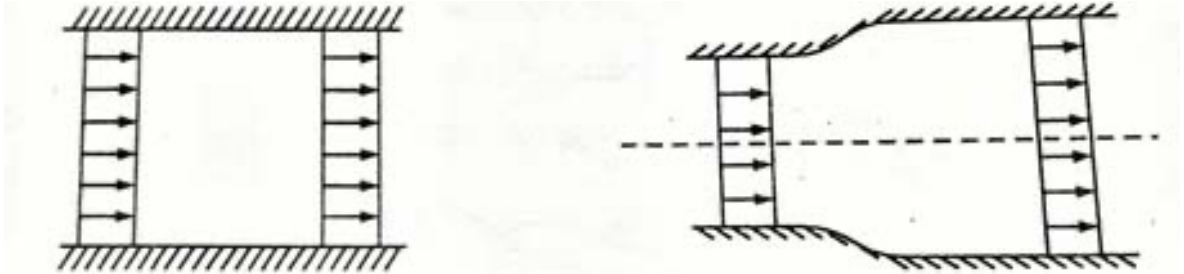
Friction ≠ o
Real Flow (μ ≠0)
Energy loss ≠ 0

One Dimensional Flow

One dimensional flow is the flow in which parameters (velocity, pressure, density, viscosity and temperature) vary only in one direction and the flow is a function of only one co-ordinate Axis and time. The flow field is represented by streamlines which are straight and parallel.

mathematically, for one-dimensional flow

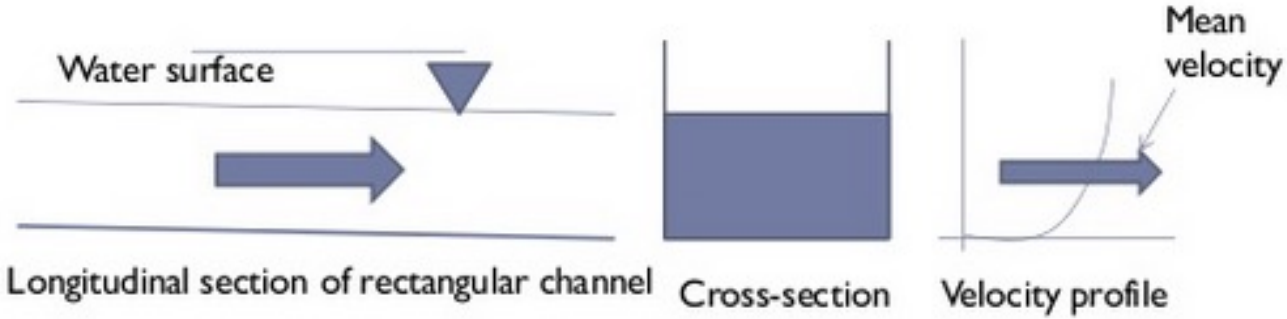
$$u = f(x),\ v = 0 \text{ and } w = 0$$



(a) Flow between parallel plates (b) incompressible flow in a duct

Illustration of one dimensional flow

- Although in general all fluids flow three-dimensionally, with pressures and velocities and other flow properties varying in all directions, in many cases the greatest changes only occur in two directions or even only in one. In these cases changes in the other direction can be effectively ignored making analysis much more simple.
- Flow is *one dimensional* *if the flow parameters (such as velocity, pressure, depth etc.) at a given instant in* time only vary in the direction of flow and not across the cross-section



Two Dimensional Flow

Two-dimensional flow is the flow in which fluid parameters vary along two directions and the flow is the function of two rectangular space coordinates (x and y-axis) and time. The flow field is represented by streamlines which are curves.

mathematically for two-dimensional flow

$$u = f_1(x, y),\ v = f_2(x, y) \text{ and } w = 0.$$

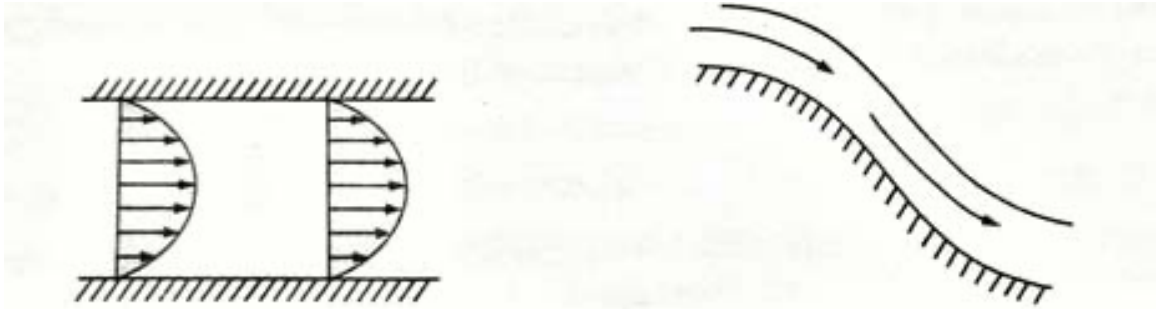


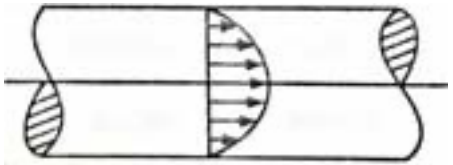
Illustration of two dimensional flow

Three Dimensional Flow

Three-dimensional flow is the flow in which flow parameters change in all the three directions and the flow is the functions of three mutually perpendicular co-ordinate Axis (x, y, z-axis) and time. The streamlines are space curves

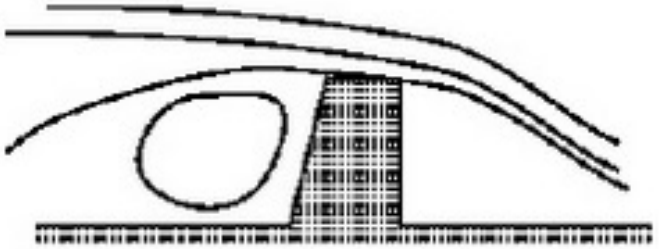
mathematically, for three-dimensional flow

$$u = f_1(x, y, z),\ v = f_2(x, y, z) \text{ and } w = f_3(x, y, z).$$



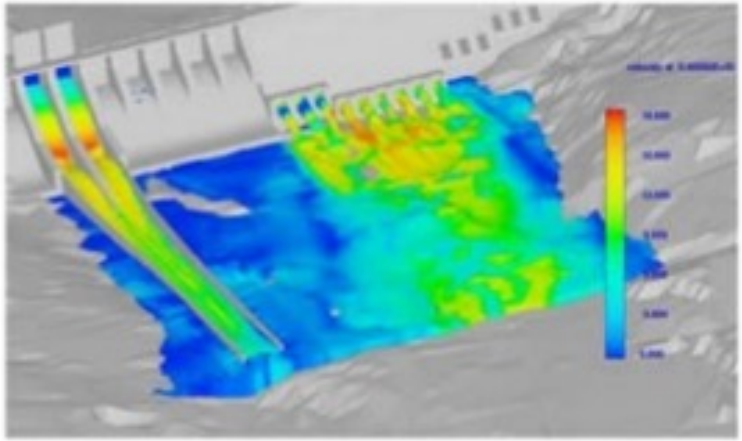
Viscous flow in a duct (three dimensional flow)

- Flow is *two-dimensional* *if it can be assumed that the flow parameters vary in the direction of flow and in* one direction at right angles to this direction



Two-dimensional flow over a weir

- Flow is *three-dimensional* *if the flow parameters vary in all three directions of flow*



Three-dimensional flow in stilling basin

Rate of Flow or Discharge

Rate of Flow or Flow Rate or Discharge:
Quantity of fluid passing through any section in a unit time.

Unit: m^3/sec

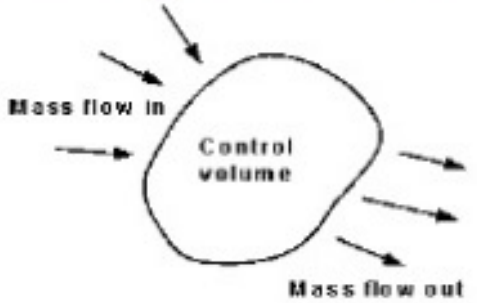
Types:

1. Volume Flow Rate = Volume of fluid / Time = Q or AV
2. Mass Flow Rate = Mass of fluid / Time = ρ x Volume / Time = ρQ or ρAV
3. Weigh Flow Rate = Weight of fluid / Time = (ρg x Volume) / Time = ρgQ or ρgAV

Where,
Q = Area x Velocity

Continuity

- Matter cannot be created or destroyed - (it is simply changed in to a different form of matter).
- This principle is know as the *conservation of mass* and we use it in the analysis of flowing fluids.
- The principle is applied to fixed volumes, known as control volumes shown in figure:



An arbitrarily shaped control volume.

For any control volume the principle of conservation of mass says

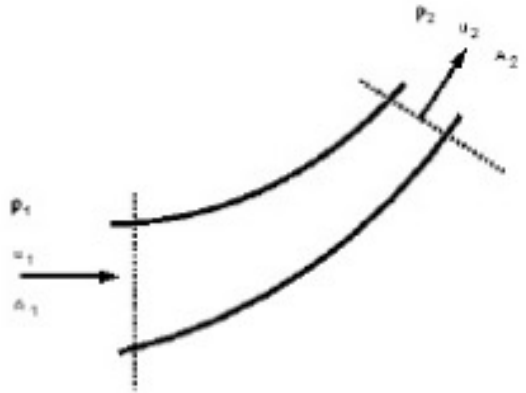
Mass entering per unit time -Mass leaving per unit time
= Increase of mass in the control volume per unit time

Continuity Equation

- For steady flow there is no increase in the mass within the control volume, so

Mass entering per unit time = Mass leaving per unit time

- **Derivation:**
- Lets consider a stream tube.
- ρ_1, v_1 and A_1 are mass density, velocity and cross-sectional area at section 1. Similarly, ρ_2, v_2 and A_2 are mass density, velocity and cross-sectional area at section 2.
- According to mass conservation



A stream tube

$$M_1 - M_2 = \frac{d(M_{CV})}{dt}$$

$$\rho_1 A_1 V_1 - \rho_2 A_2 V_2 = \frac{d(M_{CV})}{dt}$$

$$M_1 = \rho_1 A_1 V_1$$

$$M_2 = \rho_2 A_2 V_2$$

Continuity Equation

- For steady flow condition $d(M_{CV})/dt = 0$

$$\rho_1 A_1 V_1 - \rho_2 A_2 V_2 = 0 \Rightarrow \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

$$M = \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

- Hence, for stead flow condition, mass flow rate at section 1 = mass flow rate at section 2. i.e., mass flow rate is constant.
- Similarly $G = \rho_1 g A_1 V_1 = \rho_2 g A_2 V_2$

- Assuming incompressible fluid, $\rho_1 = \rho_2 = \rho$

$$A_1 V_1 = A_2 V_2 \Rightarrow Q_1 = Q_2 \Rightarrow Q_1 = Q_2 = Q_3 = Q_4$$

Continuity Equation

- Therefore, according to mass conservation for steady flow of incompressible fluids volume flow rate remains same from section to section.

VELOCITY

Velocity of Fluid Particle

- Velocity of a fluid along any direction can be defined as the rate of change of displacement of the fluid along that direction

Resultant velocity $\bar{V} = u\hat{i} + v\hat{j} + w\hat{k}$

$$|V| = \sqrt{u^2 + v^2 + w^2}$$

ACCELERATION

Acceleration of Fluid Particle

- Acceleration of a fluid element along any direction can be defined as the rate of change of velocity of the fluid along that direction.

$$\bar{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Where a_X, a_Y, a_Z are the acceleration vectors in X, Y and Z directions, respectively.

- a_x, a_y and a_z are termed as the total acceleration on respective directions.
- Total acceleration has two components – w.r.t time and w.r.t space and can be expressed in terms of u, v and w and this can be represented as:

$$a_x = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}$$

$$a_y = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}$$

$$a_z = \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}$$

local acceleration or temporal acceleration

convective acceleration

- Euler equations are applicable to compressible, incompressible, non-viscous in steady or unsteady state of flow.

Let V is the resultant velocity at any point in a fluid flow. Let u, v and w are its component in x, y and z directions. The velocity components are functions of space-co-ordinates and time. Mathematically, the velocity components are given as

$$u = f_1(x, y, z, t)$$
$$v = f_2(x, y, z, t)$$
$$w = f_3(x, y, z, t)$$

and Resultant velocity, $V = ui + vj + wk = \sqrt{u^2 + v^2 + w^2}$

Let a_x, a_y and a_z are the **total acceleration** in x, y and z directions respectively. Then by the chain rule of differentiation, we have

$$a_x = \frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\cdot\frac{dz}{dt} + \frac{\partial u}{\partial t}$$

But $\frac{dx}{dt} = u, \frac{dy}{dt} = v$ and $\frac{dz}{dt} = w$

$\therefore$ $$a_x = \frac{du}{dt} = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

Similarly, $$a_y = \frac{dv}{dt} = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = \frac{dw}{dt} = u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

...(5.6)

Local Acceleration

For steady flow, $\frac{\partial V}{\partial t} = 0$, where V is resultant velocity

or $$\frac{\partial u}{\partial t} = 0, \frac{\partial v}{\partial t} = 0 \text{ and } \frac{\partial w}{\partial t} = 0$$

Hence acceleration in x, y and z directions becomes

$$\left.\begin{aligned} a_x &= \frac{du}{dt} = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} \\ a_y &= \frac{dv}{dt} = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} \\ a_z &= \frac{dw}{dt} = u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} \end{aligned}\right\} \quad \ldots(5.7)$$

Acceleration vector $$\left.\begin{aligned} A &= a_x i + a_y j + a_z k \\ &= \sqrt{a_x^2 + a_y^2 + a_z^2}\,. \end{aligned}\right\} \quad \ldots(5.8)$$

CONTINUITY EQUATION IN THREE-DIMENSIONS

When fluid flow through a full pipe, the volume of fluid entering in to the pipe must be equal to the volume of the fluid leaving the pipe, even if the diameter of the pipe vary.

Therefore we can define the continuity equation as the equation based on the principle of conservation of mass.

Therefore, for a flowing fluid through the pipe at every cross-section, the quantity of fluid per second will be constant.

Consider a fluid element of lengths dx, dy and dz in the direction of x, y and z. Let u, v and w are the inlet velocity components in x, y and z directions respectively. Mass of fluid entering the face $ABCD$ per second

$$= \rho \times \text{Velocity in } x\text{-direction} \times \text{Area of } ABCD$$
$$= \rho \times u \times (dy \times dz)$$

Then mass of fluid leaving the face $EFGH$ per second $= \rho u\, dydz + \frac{\partial}{\partial x}(\rho u\, dydz)\, dx$

$\therefore$ Gain of mass in x-direction

$$= \text{Mass through } ABCD - \text{Mass through } EFGH \text{ per second}$$
$$= \rho u\, dydz - \rho u\, dydz - \frac{\partial}{\partial x}(\rho u\, dydz)dx$$
$$= -\frac{\partial}{\partial x}(\rho u\, dydz)\, dx$$
$$= -\frac{\partial}{\partial x}(\rho u)\, dx\, dydz \qquad \{\because dydz \text{ is constant}\}$$

Similarly, the net gain of mass in y-direction

$$= -\frac{\partial}{\partial y}(\rho v)\, dxdydz$$

and in z-direction

$$= -\frac{\partial}{\partial z}(\rho w)\, dxdydz$$

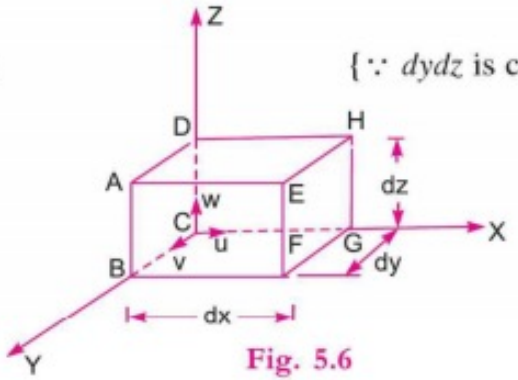


Fig. 5.6

$\therefore$ Net gain of masses $= -\left[\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w)\right] dxdydz$

Since the mass is neither created nor destroyed in the fluid element, the net increase of mass per unit time in the fluid element must be equal to the rate of increase of mass of fluid in the element. But mass of fluid in the element is $\rho.\ dx.\ dy.\ dz$ and its rate of increase with time is $\frac{\partial}{\partial t}(\rho\ dx.\ dy.\ dz)$ or

$$\frac{\partial \rho}{\partial t}\ .\ dx\ dy\ dz.$$

Equating the two expressions,

or $$-\left[\frac{\partial}{\partial x}(\rho u)+\frac{\partial}{\partial y}(\rho v)+\frac{\partial}{\partial z}(\rho w)\right] dxdydz = \frac{\partial \rho}{\partial t}.\ dxdydz$$

or $$\frac{\partial \rho}{\partial t}+\frac{\partial}{\partial x}(\rho u)+\frac{\partial}{\partial y}(\rho v)+\frac{\partial}{\partial z}(\rho w) = 0$$ [Cancelling $dx.dy.dz$ from both sides] ...(5.3A)

Integration of both sides gives $\int dw = \int\left(-3x - 2xy + z^2\right) dz$

or $$w = \left(-3xz - 2xyz + \frac{z^3}{3}\right) + \text{Constant of integration,}$$

where constant of integration cannot be a function of z. But it can be a function of x and y that is $f(x, y)$.

$\therefore$ $$w = \left(\mathbf{-3xz - 2xyz + \frac{z^3}{3}}\right) \mathbf{+ f(x, y).\ Ans.}$$

Case II. $$v = 2y^2 \quad \therefore \quad \frac{\partial v}{\partial y} = 4y$$

$$w = 2xyz \quad \therefore \quad \frac{\partial w}{\partial z} = 2xy$$

Substituting the values of $\frac{\partial v}{\partial y}$ and $\frac{\partial w}{\partial z}$ in continuity equation, we get

$$\frac{\partial u}{\partial x} + 4y + 2xy = 0$$

or $$\frac{\partial u}{\partial x} = -4y - 2xy \text{ or } du = (-4y - 2xy)\, dx$$

Integrating, we get $$u = -4xy - 2y\frac{x^2}{2} + f(y, z) = \mathbf{-4xy - x^2y + f(y, z).\ Ans.}$$

Stream Function and Velocity Potential Function

5.8.1 Velocity Potential Function. It is defined as a scalar function of space and time such that its negative derivative with respect to any direction gives the fluid velocity in that direction. It is defined by ϕ (Phi). Mathematically, the velocity, potential is defined as $\phi = f(x, y, z)$ for steady flow such that

$$\left.\begin{aligned} u &= -\frac{\partial \phi}{\partial x} \\ v &= -\frac{\partial \phi}{\partial y} \\ w &= -\frac{\partial \phi}{\partial z} \end{aligned}\right\} \quad \ldots(5.9)$$

where u, v and w are the components of velocity in x, y and z directions respectively.

The continuity equation for an incompressible steady flow is $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0.$

Substituting the values of u, v and w from equation (5.9), we get

$$\frac{\partial}{\partial x}\left(-\frac{\partial \phi}{\partial x}\right)+\frac{\partial}{\partial y}\left(-\frac{\partial \phi}{\partial y}\right)+\frac{\partial}{\partial z}\left(-\frac{\partial \phi}{\partial z}\right)=0$$

or
$$\frac{\partial^2 \phi}{\partial x^2}+\frac{\partial^2 \phi}{\partial y^2}+\frac{\partial^2 \phi}{\partial z^2}=0. \qquad ...(5.10)$$

Equation (5.10) is a Laplace equation.

For two-dimension case, equation (5.10) reduces to $\frac{\partial^2 \phi}{\partial x^2}+\frac{\partial^2 \phi}{\partial y^2}=0.$ ...(5.11)

If any value of ϕ that satisfies the Laplace equation, will correspond to some case of fluid flow.

Properties of the Potential Function. The rotational components* are given by

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$\omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)$$

Substituting the values, of u, v and w from equation (5.9) in the above rotational components, we get

$$\omega_z = \frac{1}{2}\left[\frac{\partial}{\partial x}\left(-\frac{\partial \phi}{\partial y}\right) - \frac{\partial}{\partial y}\left(-\frac{\partial \phi}{\partial x}\right)\right] = \frac{1}{2}\left[-\frac{\partial^2 \phi}{\partial x \partial y} + \frac{\partial^2 \phi}{\partial y \partial x}\right]$$

$$\omega_y = \frac{1}{2}\left[\frac{\partial}{\partial z}\left(-\frac{\partial \phi}{\partial x}\right) - \frac{\partial}{\partial x}\left(-\frac{\partial \phi}{\partial z}\right)\right] = \frac{1}{2}\left[-\frac{\partial^2 \phi}{\partial z \partial x} + \frac{\partial^2 \phi}{\partial x \partial z}\right]$$

and

$$\omega_x = \frac{1}{2}\left[\frac{\partial}{\partial y}\left(-\frac{\partial \phi}{\partial z}\right) - \frac{\partial}{\partial z}\left(-\frac{\partial \phi}{\partial y}\right)\right] = \frac{1}{2}\left[-\frac{\partial^2 \phi}{\partial y \partial z} + \frac{\partial^2 \phi}{\partial z \partial y}\right]$$

If ϕ is a continuous function, then $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$; $\frac{\partial^2 \phi}{\partial z \partial x} = \frac{\partial^2 \phi}{\partial x \partial z}$; etc.

$$\therefore \qquad \omega_z = \omega_y = \omega_x = 0.$$

When rotational components are zero, the flow is called irrotational. Hence the properties of the potential function are :

1. If velocity potential (ϕ) exists, the flow should be irrotational.

2. If velocity potential (ϕ) satisfies the Laplace equation, it represents the possible steady incompressible irrotational flow.

Problem 5.11 *The velocity potential function is given by* $\phi = 5\,(x^2 - y^2)$. *Calculate the velocity components at the point (4, 5).*

Solution.

$$\phi = 5\,(x^2 - y^2)$$

$$\therefore \quad \frac{\partial \phi}{\partial x} = 10x$$

$$\frac{\partial \phi}{\partial y} = -\,10y.$$

But velocity components u and v are given by equation (5.9) as

$$u = -\frac{\partial \phi}{\partial x} = -\,10x$$

$$v = -\frac{\partial \phi}{\partial y} = -\,(-\,10y) = 10y$$

The velocity components at the point (4, 5), *i.e.*, at $x = 4$, $y = 5$

$u = -\,10 \times 4 =$ **$-$ 40 units. Ans.**

$v = 10 \times 5 =$ **50 units. Ans.**

5.8.2 Stream Function. It is defined as the scalar function of space and time, such that its partial derivative with respect to any direction gives the velocity component at right angles to that direction. It is denoted by ψ (*Psi*) and defined only for two-dimensional flow. Mathematically, for steady flow it is defined as $\psi = f(x, y)$ such that

$$\left.\begin{aligned} \frac{\partial \psi}{\partial x} &= v \\ \text{and} \quad \frac{\partial \psi}{\partial y} &= -u \end{aligned}\right\} \qquad \ldots(5.12)$$

The velocity components in cylindrical polar co-ordinates in terms of stream function are given as

$$u_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} \text{ and } u_\theta = -\frac{\partial \psi}{\partial r} \qquad \ldots(5.12A)$$

where u_r= radial velocity and u_θ = tangential velocity

The continuity equation for two-dimensional flow is $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$.

Substituting the values of u and v from equation (5.12), we get

$$\frac{\partial}{\partial x}\left(-\frac{\partial \psi}{\partial y}\right)+\frac{\partial}{\partial y}\left(\frac{\partial \psi}{\partial x}\right)=0 \text{ or } -\frac{\partial^2 \psi}{\partial x \partial y}+\frac{\partial^2 \psi}{\partial x \partial y}=0.$$

Hence existence of ψ means a possible case of fluid flow. The flow may be rotational or irrotational.

The rotational component ω_z is given by $\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)$.

Substituting the values of u and v from equation (5.12) in the above rotational component, we get

$$\omega_z = \frac{1}{2}\left[\frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial x}\right)-\frac{\partial}{\partial y}\left(-\frac{\partial \psi}{\partial y}\right)\right]=\frac{1}{2}\left[\frac{\partial^2 \psi}{\partial x^2}+\frac{\partial^2 \psi}{\partial y^2}\right]$$

For irrotational flow, $\omega_z = 0$. Hence above equation becomes as $\frac{\partial^2 \psi}{\partial x^2}+\frac{\partial \psi}{\partial y^2}=0$

which is Laplace equation for ψ.

The **properties** of stream function (ψ) are :

1. If stream function (ψ) exists, it is a possible case of fluid flow which may be rotational or irrotational.
2. If stream function (ψ) satisfies the Laplace equation, it is a possible case of an irrotational flow.

5.8.6 Relation between Stream Function and Velocity Potential Function

From equation (5.9),

we have $$u = -\frac{\partial \phi}{\partial x} \text{ and } v = -\frac{\partial \phi}{\partial y}$$

From equation (5.12), we have $u = -\frac{\partial \psi}{\partial y}$ and $v = \frac{\partial \psi}{\partial x}$

Thus, we have $$u = -\frac{\partial \phi}{\partial x} = -\frac{\partial \psi}{\partial y} \text{ and } v = -\frac{\partial \phi}{\partial y} = \frac{\partial \psi}{\partial x}$$

Hence $$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$$

and $$\frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

...(5.15)

Energy Equation

Newton Second law of motion : Rate of change of momentum is equal to forces action on body,

$$\vec{F} = \left.\frac{d\vec{P}}{dt}\right)_{\text{system}}$$

where the linear momentum, $\vec{P}$, of the system is given by

$$\vec{P}_{\text{system}} = \int_{\text{mass (system)}} \vec{V}\, dm$$

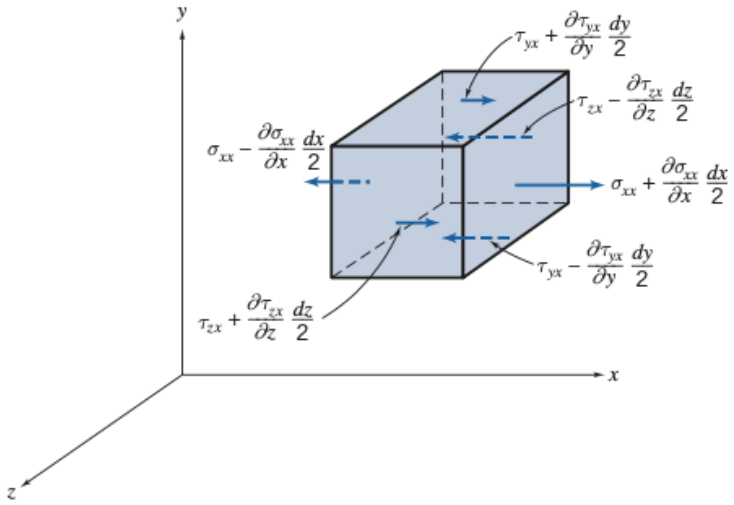
For an infinitesimal system of mass dm, Newton's second law can be written,

$$d\vec{F} = dm \left.\frac{d\vec{V}}{dt}\right)_{\text{system}}$$

$$d\vec{F} = dm \frac{D\vec{V}}{Dt} = dm \left[u\frac{\partial \vec{V}}{\partial x} + v\frac{\partial \vec{V}}{\partial y} + w\frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t} \right] \quad (1)$$

Forces Acting on Particle

- Body Force
- Surface Force
  - Normal Force
  - Tangential Force

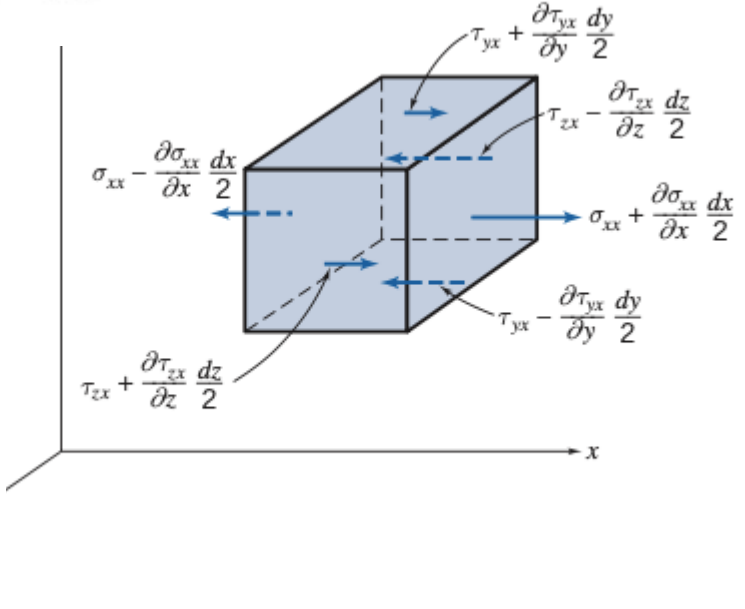


$$dF_{S_x} = \left(\sigma_{xx} + \frac{\partial \sigma_{xx}}{\partial x}\frac{dx}{2}\right)dy\,dz - \left(\sigma_{xx} - \frac{\partial \sigma_{xx}}{\partial x}\frac{dx}{2}\right)dy\,dz$$
$$+ \left(\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y}\frac{dy}{2}\right)dx\,dz - \left(\tau_{yx} - \frac{\partial \tau_{yx}}{\partial y}\frac{dy}{2}\right)dx\,dz$$
$$+ \left(\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z}\frac{dz}{2}\right)dx\,dy - \left(\tau_{zx} - \frac{\partial \tau_{zx}}{\partial z}\frac{dz}{2}\right)dx\,dy$$

$$dF_{S_x} = \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right)dx\,dy\,dz$$

$$dF_y = dF_{B_y} + dF_{S_y} = \left(\rho g_y + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z}\right)dx\,dy\,dz$$

$$dF_z = dF_{B_z} + dF_{S_z} = \left(\rho g_z + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z}\right)dx\,dy\,dz$$



If we substitute these expressions for the force components into the x, y, and z components of Eq. (1), we obtain the differential equations of motion (),

$$\rho g_x + \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} = \rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right) \qquad (2)$$

$$\rho g_y + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} = \rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right) \qquad (3)$$

$$\rho g_z + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho\left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right) \qquad (4)$$

Above equations are the differential equations of motion for any fluid satisfying the continuum assumption.

Navier Stokes Equations

For a Newtonian fluid the viscous stress is directly proportional to the rate of shearing strain
(angular deformation rate).
One-dimensional laminar Newtonian flow the shear stress is proportional to the rate of angular deformation

$$\tau_{y,x} = \mu \times \frac{\partial u}{\partial y}$$

For a three-dimensional flow the equation is expressed as:

$$\tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)$$

$$\tau_{yz} = \tau_{zy} = \mu\left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)$$

$$\tau_{zx} = \tau_{xz} = \mu\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)$$

Similarly, Normal stress $\sigma_{xx}, \sigma_{yy}, \& \sigma_{zz}$ are expressed:

$$\sigma_{xx} = -p - \frac{2}{3}\mu\nabla\cdot\vec{V} + 2\mu\frac{\partial u}{\partial x}$$

$$\sigma_{yy} = -p - \frac{2}{3}\mu\nabla\cdot\vec{V} + 2\mu\frac{\partial v}{\partial y}$$

$$\sigma_{zz} = -p - \frac{2}{3}\mu\nabla\cdot\vec{V} + 2\mu\frac{\partial w}{\partial z}$$

Where p is pressure

Substituting value of $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \tau_{y,x}, \tau_{y,z}$ & $\tau_{z,x}$ in equation (2), (3) and (4), we will get

$$\rho\frac{Du}{Dt} = \rho g_x - \frac{\partial p}{\partial x} + \frac{\partial}{\partial x}\left[\mu\left(2\frac{\partial u}{\partial x} - \frac{2}{3}\nabla\cdot\vec{V}\right)\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right)\right]$$

$$\rho\frac{Dv}{Dt} = \rho g_y - \frac{\partial p}{\partial y} + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\right] + \frac{\partial}{\partial y}\left[\mu\left(2\frac{\partial v}{\partial y} - \frac{2}{3}\nabla\cdot\vec{V}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right)\right]$$

$$\rho\frac{Dw}{Dt} = \rho g_z - \frac{\partial p}{\partial z} + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right)\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial w}{\partial z} + \frac{\partial w}{\partial y}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(2\frac{\partial w}{\partial z} - \frac{2}{3}\nabla\cdot\vec{V}\right)\right]$$

These equations of motion are called the Navier stokes equations. The equations are greatly simplified when applied to incompressible flow with constant viscosity.

For incompressible and Newtonian fluid, Navier stokes equation can be expressed as:

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=\rho g_x-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)$$

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=\rho g_y-\frac{\partial p}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right)$$

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=\rho g_z-\frac{\partial p}{\partial z}+\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right)$$

In the fluid flow, the following forces are present :

(*i*) F_g, gravity force.

(*ii*) F_p, the pressure force.

(*iii*) F_v, force due to viscosity.

(*iv*) F_t, force due to turbulence.

(*v*) F_c, force due to compressibility.

Thus in equation (6.1), the net force

$$F_x = (F_g)_x + (F_p)_x + (F_v)_x + (F_t)_x + (F_c)_x.$$

(*i*) If the force due to compressibility, F_c is negligible, the resulting net force

$$F_x = (F_g)_x + (F_p)_x + (F_v)_x + (F_t)_x$$

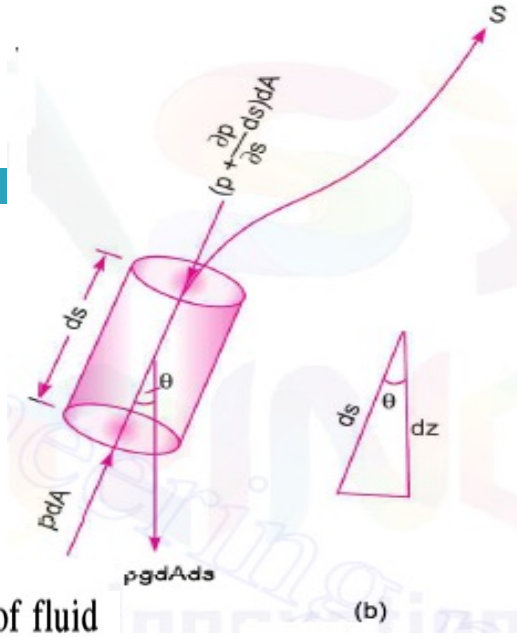
and equation of motions are called **Reynold's equations of motion.**

(*ii*) For flow, where (F_t) is negligible, the resulting equations of motion are known as **Navier-Stokes Equation.**

(*iii*) If the flow is assumed to be ideal, viscous force (F_v) is zero and equation of motions are known as **Euler's equation of motion.**

Energy Equation

1. Pressure force pdA in the direction of flow.
2. Pressure force $\left(p + \frac{\partial p}{\partial s} ds\right) dA$ opposite to the direction of flow.
3. Weight of element $\rho g dA ds$.



The resultant force on the fluid element in the direction of s must be equal to the mass of fluid element × acceleration in the direction s.

$$\therefore \quad pdA - \left(p + \frac{\partial p}{\partial s} ds\right) dA - \rho g dA ds \cos\theta$$

$$= \rho dA ds \times a_s$$

where a_s is the acceleration in the direction of s.

Now $a_s = \frac{dv}{dt}$, where v is a function of s and t.

$$= \frac{\partial v}{\partial s}\frac{ds}{dt} + \frac{\partial v}{\partial t} = \frac{v\partial v}{\partial s} + \frac{\partial v}{\partial t} \quad \left\{\because \frac{ds}{dt} = v\right\}$$

If the flow is steady, $\frac{\partial v}{\partial t} = 0$

$$\therefore \quad a_s = \frac{v\partial v}{\partial s}$$

Substituting the value of a_s in equation (6.2) and simplifying the equation, we get

$$-\frac{\partial p}{\partial s}\, dsdA - \rho g\, dAds \cos\theta = \rho dAds \times \frac{\partial v}{\partial s}$$

Dividing by $\rho dsdA$, $-\frac{\partial p}{\rho \partial s} - g\cos\theta = \frac{v\partial v}{\partial s}$

or $$\frac{\partial p}{\rho \partial s} + g\cos\theta + v\frac{\partial v}{\partial s} = 0$$

we have $\cos\theta = \frac{dz}{ds}$

$$\therefore \quad \frac{1}{\rho}\frac{dp}{ds} + g\frac{dz}{ds} + \frac{vdv}{ds} = 0 \quad \text{or} \quad \frac{dp}{\rho} + gdz + vdv = 0$$

or $$\frac{dp}{\rho} + gdz + vdv = 0$$

Bernoulli Equation

Bernoulli's equation is obtained by integrating the Euler's equation of motion

$$\int \frac{dp}{\rho} + \int g dz + \int v dv = \text{constant}$$

If flow is incompressible, ρ is constant and

$$\therefore \quad \frac{p}{\rho} + gz + \frac{v^2}{2} = \text{constant}$$

or

$$\frac{p}{\rho g} + z + \frac{v^2}{2g} = \text{constant}$$

or

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$$

$\frac{p}{\rho g}$ = pressure energy per unit weight of fluid or pressure head.

$v^2/2g$ = kinetic energy per unit weight or kinetic head.

z = potential energy per unit weight or potential head.

Assumption :

- Fluid is ideal , (Viscosity = 0)
- Flow is steady
- Flow is irrotational
- Flow is incompressible

Momentum Equation

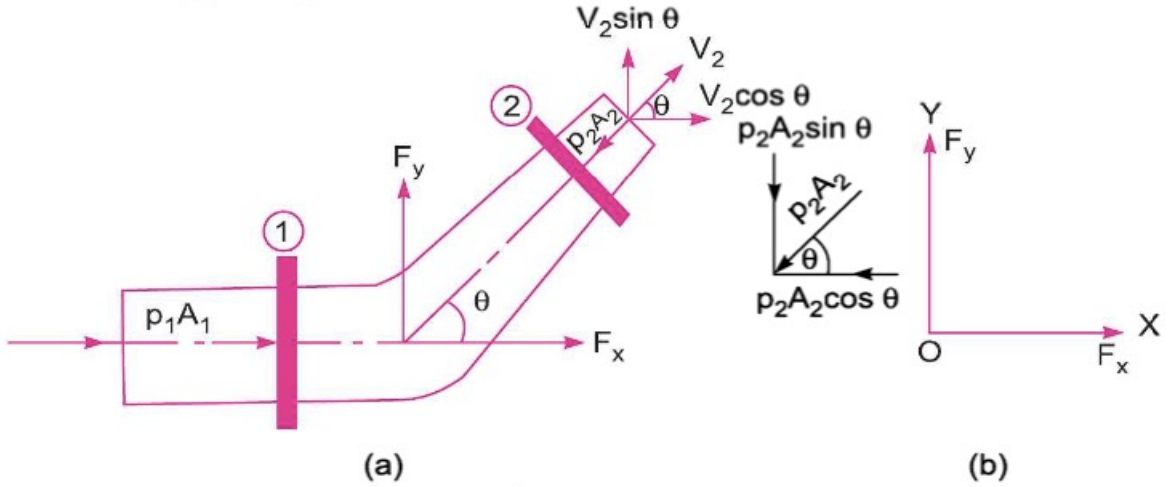
Consider two sections (1) and (2), as shown in Fig.

Let v_1 = velocity of flow at section (1),

p_1 = pressure intensity at section (1),

A_1 = area of cross-section of pipe at section (1) and

v_2, p_2, A_2 = corresponding values of velocity, pressure and area at section (2).



Forces on bend.

Net force acting on fluid in the direction of x = Rate of change of momentum in x-direction

$$\therefore \quad p_1A_1 - p_2A_2 \cos\theta - F_x = \text{(Mass per sec) (change of velocity)}$$

$$= \rho Q \text{ (Final velocity in the direction of } x$$

$$- \text{Initial velocity in the direction of } x)$$

$$= \rho Q\,(V_2 \cos\theta - V_1)$$

$$\therefore \quad F_x = \rho Q\,(V_1 - V_2 \cos\theta) + p_1A_1 - p_2A_2 \cos\theta$$

Similarly the momentum equation in y-direction gives

$$0 - p_2A_2 \sin\theta - F_y = \rho Q\,(V_2 \sin\theta - 0)$$

$$\therefore \quad F_y = \rho Q\,(-V_2 \sin\theta) - p_2A_2 \sin\theta$$

Now the resultant force (F_R) acting on the bend

$$= \sqrt{F_x^2 + F_y^2}$$

And the angle made by the resultant force with horizontal direction is given by

$$\tan\theta = \frac{F_y}{F_x}$$

Moment of Momentum Equation

It is derived from moment of momentum principal

Momentum Principal – Resulting torque acting on rotating fluid is equal to the rate of change of momentum

Let V_1 = velocity of fluid at section 1,

r_1 = radius of curvature at section 1,

Q = rate of flow of fluid,

ρ = density of fluid,

and V_2 and r_2 = velocity and radius of curvature at section 2

Momentum of fluid at section 1 = mass × velocity = $\rho Q \times V_1/s$

$\therefore$ Moment of momentum per second at section 1,

$= \rho Q \times V_1 \times r_1$

Similarly moment of momentum per second of fluid at section 2

$$= \rho Q \times V_2 \times r_2$$

$\therefore$ Rate of change of moment of momentum

$$= \rho Q V_2 r_2 - \rho Q V_1 r_1 = \rho Q [V_2 r_2 - V_1 r_1]$$

According to moment of momentum principle

Resultant torque = rate of change of moment of momentum

or
$$T = \rho Q [V_2 r_2 - V_1 r_1]$$

Equation is known as moment of momentum equation. This equation is applied :

1. For analysis of flow problems in turbines and centrifugal pumps.
2. For finding torque exerted by water on sprinkler.